

Ruoqing Feng

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RESEARCH PROFILE

I am interested in understanding how cellular systems reorganize during aging and neurodegeneration, particularly through neuroimmune interactions, inflammatory remodeling, and epigenetic regulation. My research combines large-scale genomics and computational analysis to study how cell states, spatial organization, and intercellular communication emerge across complex tissue environments. I am especially drawn to unbiased, data-driven approaches that integrate multiple layers of omics information to uncover regulatory principles underlying cellular transitions in the aging and diseased brain.

EDUCATIONAL BACKGROUND

German Center for Neurodegenerative Diseases (LMU-GSN)

PhD Student in the lab of Prof. Mikael Simons.

03/2023 – Present

Southern University of Science and Technology

Master of Science in Biology

09/2019 – 06/2022

Southern University of Science and Technology

Bachelor of Science in Bioinformatics

09/2015 – 06/2019

WORKING EXPERIENCE

Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences

Research assistant in Bioinformatics

06/2022 – 03/2023

PUBLICATIONS

Publications as co-first author

Feng, R., Spieth, L., Liu, L., Simons, M., et al. Single-cell spatial transcriptomic profiling defines a pathogenic inflammatory niche in chronic active multiple sclerosis lesions. *Immunity*. 2025 Dec 9;58(12):2989-3005.e10. <https://doi.org/10.1016/j.immuni.2025.10.003>

Wu, Z., Wang, Z., Wang, H., **Feng, R.**, Li, H., et al. An immune cell atlas reveals the dynamics of human macrophage specification during prenatal development. *Cell*, Volume 186, Issue 20, 4454 - 4471.e19. <https://doi.org/10.1016/j.cell.2023.08.019>

Publications as co-author

Groh, J., **Feng, R.**, Yuan, X. et al. Microglia activation orchestrates CXCL10-mediated CD8+ T cell recruitment to promote aging-related white matter degeneration. *Nat Neurosci* 28, 1160–1173 (2025). <https://doi.org/10.1038/s41593-025-01955-w>

Kedia, S., Ji, H., **Feng, R.** et al. T cell-mediated microglial activation triggers myelin pathology in a mouse model of amyloidosis. *Nat Neurosci* 27, 1468–1474 (2024). <https://doi.org/10.1038/s41593-024-01682-8>

Zhao, Y., Sun, J., Li, Y., Li, Z., Xie, Y., **Feng, R.**, Zhao, J., & Hu, Y. (2021). The strand-biased transcription of SARS-CoV-2 and unbalanced inhibition by remdesivir. *IScience*, 24(8), 102857. <https://doi.org/10.1016/j.isci.2021.102857>

RESEARCH EXPERIENCE

A single-cell resolved spatial transcriptomics atlas of human multiple sclerosis lesion

05/2023 – Present

- Led the full computational analysis and data integration of spatial transcriptomic and single-cell datasets from human multiple sclerosis lesions.
- Developed the cell-type annotation strategy and integrated single-cell reference information to resolve glial, immune, and lesion-associated cellular states.
- Performed brain-region and lesion-area segmentation to define spatially organized pathological structures within chronic active lesions.
- Established the analytical framework for defining inflammatory niches by integrating regional segmentation, spatial neighborhood composition, and cell-type distribution patterns.

- Conducted cell–cell interaction analyses to identify candidate communication programs between microglia, infiltrating immune cells, and other lesion-associated cell populations.
- Proposed downstream experimental validation strategies to test computationally inferred mechanisms, including neuroimmune interactions and inflammatory niche organization.

A single cell RNA-seq atlas of the human developing immune system

11/2020 – 03/2023

- Led the computational integration and annotation of large-scale single-cell RNA-seq datasets across human developmental tissues.
- Established the immune-cell annotation framework, including systematic naming and classification of major immune populations and tissue-associated macrophage subsets.
- Designed and implemented trajectory analysis strategies to reconstruct developmental relationships among macrophage and immune-cell states.
- Characterized tissue-specific immune-cell programs and identified developmental patterns underlying macrophage specification across prenatal organs.
- Contributed to the biological interpretation of proangiogenic macrophage states, perivascular localization, and developmental immune-cell diversification.
- Helped define analysis concepts linking cell-state transitions, tissue context, and developmental trajectories in the human immune system.

SELECTED PRESENTATIONS & POSTERS

FENS-Hertie Winter School 2025-2026 | Flash Talk

01/2026

Spatial and single-cell transcriptomic analysis of inflammatory niches in multiple sclerosis lesions

GSN Students' Conference | Presentation

03/2026

Mapping MS lesions with spatial transcriptomic

Brain (epi)genome | Poster (Best Poster Prize)

04/2026

Spatial and single-cell transcriptomic analysis of inflammatory niches in multiple sclerosis lesions

TECHNICAL EXPERTISE

Spatial and single-cell omics

MERFISH, Visium, Stereo-seq, scRNA-seq, snRNA-seq, scATAC-seq, multimodal integration, reference-based annotation, spatial niche analysis.

Computational genomics

RNA-seq analysis, differential expression, gene set enrichment, trajectory inference, cell–cell communication, gene regulatory network analysis.

Programming and tools

R, Python, Linux/HPC.

Wet-lab techniques

MERFISH tissue preparation and imaging, Western blotting.